

## tests/graphCityBridge.test.ts

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### Overview

A new test suite `tests/graphCityBridge.test.ts` has been added to validate the mapping utilities in `graphCityBridge`. The file imports `Vitest`, `KnowledgeGraph`, and several helper functions (`cityBuildingIdsForHighlight`, `cityBuildingToGraphNodeId`, `graphEdgesToCityPairs`, `graphNodeToCityBuildingIds`) along with `CodeCity` layout types.

### Key changes

- **Test file added:** `tests/graphCityBridge.test.ts` (≈121 lines).
- **Imports:** `vitest` helpers, `KnowledgeGraph` type, and the four mapping functions from `../src/app/graphCityBridge`.
- **Helper `building()`:** constructs `CodeCityRenderableBuilding` objects with default geometry and overrides.
- **Sample layout & graph:** a minimal `CodeCityLayoutResult` with two buildings (function `foo` and class `Bar`) and a corresponding `KnowledgeGraph` containing nodes and a single `calls` edge.
- **Four test cases:**
  1. `graphNodeToCityBuildingIds` maps a function node to its city building ID.
  2. `cityBuildingToGraphNodeId` reverses the mapping.
  3. `cityBuildingIdsForHighlight` returns all building IDs for a file node.
  4. `graphEdgesToCityPairs` produces overlay pairs for a selected node.

### Impact

- **Correctness:** Provides concrete assertions for the mapping logic, reducing silent regressions.
- **Maintainability:** Centralizes test data; future changes to node/building IDs must update this file.
- **Observability:** Test failures will surface mapping mismatches immediately during CI.
- **Compatibility:** No changes to production code; only test additions.

### Risks & follow-ups

- **Regression risk:** If `graphCityBridge` functions change signatures or ID conventions, tests will fail; verify that the mapping logic remains stable.
- **Snapshot drift:** No snapshots used, but any change to default geometry in `building()` may require test updates.
- **Linting:** Ensure the new file passes the repo's TS lint rules (`tsc`, `eslint`).
- **Build impact:** Run `npm run build` to confirm the added test does not introduce build-time errors.